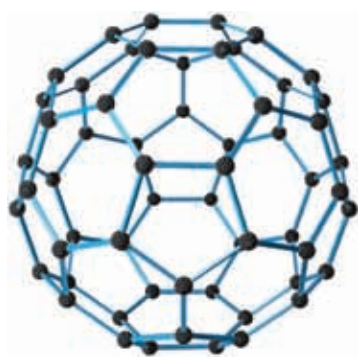


Nanoparticles are of interest because of the new properties (such as chemical reactivity and optical behaviour) that they exhibit compared with larger particles of the same materials. For example, titanium dioxide and zinc oxide become transparent at the nanoscale and are able to absorb and reflect UV light, and have found application in sunscreens. The bending of bulk copper (wire, ribbon, etc.) occurs with movement of copper atoms/clusters at about the 50 nm scale. Copper nanoparticles smaller than 50 nm are considered super hard materials that do not exhibit the same malleability and ductility as bulk copper.

In the short-term, nanoparticles can be used in new cosmetics, textiles and paints and in the longer term, it has its applications in methods of targeted drug delivery where they could be used to deliver drugs to a specific site in the body.

Fullerenes (Buckyballs)

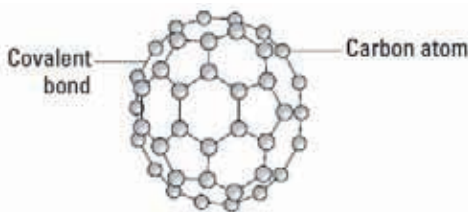
Buckyballs were discovered through an interesting collaboration of researchers from two universities. Richard Smalley at Rice University was studying semiconductor materials. He had a device that shined a laser at a solid sample, vaporized part of it, and analyzed the clusters of atoms that formed in the vapor. Meanwhile, at the University of Sussex, Harry Kroto as well as Bob Curl were attempting to reproduce a material found in deep space that



Buckyballs

generated specific molecular spectra from carbon atoms. Bob Curl, also from

Rice University, was doing similar work. When Kroto dropped by Rice, and saw the work that Smalley was doing, he became interested in using that equipment to reproduce his carbon molecules.. In August of 1985, Smalley, Kroto,



Carbon Bonding in Buckyballs

Curl, and some graduate students performed a series of experiments producing carbon molecules and clusters. They found that under certain conditions, most of the molecules generated contained 60 carbon atoms.